

The book shows how the central limit theorem for independent, identically distributed random variables with values in general, multidimensional spaces holds uniformly over some large classes of functions.

The book contains, with complete proofs, the Fernique-Talagrand majorizing measure theorem for Gaussian processes, an extended treatment of Vapnik-Černovenkis combinatorics, the Ossiander L^2 bracketing central limit theorem, the Giné-Zinn bootstrap central limit theorem in probability, the Bronstein theorem on approximation of convex sets, and the Shor theorem on rates of convergence over lower layers. The book incorporates an updated form of the author's 1984 St.-Flour lecture notes and also gives various results of the author's not previously collected in one place. A number of recent results of Talagrand and others are surveyed without proofs in separate sections.

The book will interest mathematicians working in probability, mathematical statisticians, and computer scientists working in computer learning theory.

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